

DBMS Skill Task – Week 6

Product Review and Rating Management System

1. REVIEW TABLE:

Purpose: Stores the customer's written feedback for a product.

Attribute	Data Type	Constraint
REVIEW_ID	NUMBER	Primary Key
CUSTOMER_ID	NUMBER	Foreign Key
PRODUCT_ID	NUMBER	Foreign Key
REVIEW_TEXT	VARCHAR2(500)	NOT NULL
REVIEW_DATE	DATE	NOT NULL

2. RATING TABLE:

Purpose: Stores the rating given by the customer.

Attribute	Data Type	Constraint
RATING_ID	NUMBER	Primary Key
CUSTOMER_ID	NUMBER	Foreign Key
PRODUCT_ID	NUMBER	Foreign Key
RATING	NUMBER(1)	CHECK (1–5)
RATING_DATE	DATE	NOT NULL

3.CREATE REVIEW TABLE:

```
SQL> CREATE TABLE REVIEW (  
 2     REVIEW_ID NUMBER PRIMARY KEY,  
 3     CUSTOMER_ID NUMBER NOT NULL,  
 4     PRODUCT_ID NUMBER NOT NULL,  
 5     REVIEW_TEXT VARCHAR2(500) NOT NULL,  
 6     REVIEW_DATE DATE NOT NULL  
 7 );
```

Table created.

4.CREATE RATING TABLE:

```
SQL> CREATE TABLE RATING (  
 2     RATING_ID NUMBER PRIMARY KEY,  
 3     CUSTOMER_ID NUMBER NOT NULL,  
 4     PRODUCT_ID NUMBER NOT NULL,  
 5     RATING NUMBER(1) CHECK (RATING BETWEEN 1 AND 5),  
 6     RATING_DATE DATE NOT NULL  
 7 );
```

Table created.

5.INSERTING REVIEW VALUES:

```
SQL> INSERT INTO REVIEW (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
 2 VALUES (11, 111, 207, 'Excellent product and very useful', SYSDATE);
```

1 row created.

SQL>

```
SQL> INSERT INTO REVIEW (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
 2 VALUES (12, 112, 207, 'Good product and worth the price', SYSDATE);
```

1 row created.

SQL>

```
SQL> INSERT INTO REVIEW (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
 2 VALUES (13, 113, 208, 'Amazing quality and fast delivery', SYSDATE);
```

1 row created.

SQL>

```
SQL> INSERT INTO REVIEW (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
 2 VALUES (14, 114, 208, 'Good quality product', SYSDATE);
```

1 row created.

SQL>

```
SQL> INSERT INTO REVIEW (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
 2 VALUES (15, 115, 209, 'Product is okay but could be better', SYSDATE);
```

1 row created.

SQL>

```
SQL> COMMIT;
```

6.INSERTING RATINGS VALUES:

```
SQL> INSERT INTO RATING (RATING_ID, CUSTOMER_ID, PRODUCT_ID, RATING, RATING_DATE)
  2  VALUES (16, 116, 210, 5, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO RATING (RATING_ID, CUSTOMER_ID, PRODUCT_ID, RATING, RATING_DATE)
  2  VALUES (17, 117, 210, 4, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO RATING (RATING_ID, CUSTOMER_ID, PRODUCT_ID, RATING, RATING_DATE)
  2  VALUES (18, 118, 211, 5, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO RATING (RATING_ID, CUSTOMER_ID, PRODUCT_ID, RATING, RATING_DATE)
  2  VALUES (19, 119, 211, 4, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO RATING (RATING_ID, CUSTOMER_ID, PRODUCT_ID, RATING, RATING_DATE)
  2  VALUES (20, 120, 212, 3, SYSDATE);

1 row created.

SQL>
SQL> COMMIT;
```

7.RETRIEVE PRODUCT REVIEW DETAILS:

```
SQL> SELECT
  2     R.REVIEW_ID,
  3     R.CUSTOMER_ID,
  4     R.PRODUCT_ID,
  5     P.PRODUCT_NAME,
  6     R.REVIEW_TEXT,
  7     R.REVIEW_DATE
  8 FROM REVIEW R
  9 JOIN PRODUCT P
 10 ON R.PRODUCT_ID = P.PRODUCT_ID
 11 ORDER BY R.REVIEW_DATE DESC;

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
          3          108          205
Shoes
Amazing quality and fast delivery
07-OCT-26
```

8. CALCULATE AVERAGE PRODUCT RATINGS:

```
SQL> SELECT
  2     PRODUCT_ID,
  3     ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  4 FROM RATING
  5 GROUP BY PRODUCT_ID
  6 ORDER BY AVERAGE_RATING DESC;
```

PRODUCT_ID	AVERAGE_RATING
211	4.5
202	4.5
201	4.5
210	4.5
212	3
203	3

6 rows selected.

9.IDENTIFY HIGHLY RATED PRODUCTS:

```
SQL> SELECT
  2     P.PRODUCT_ID,
  3     P.PRODUCT_NAME,
  4     ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5 FROM RATING R
  6 JOIN PRODUCT P
  7 ON R.PRODUCT_ID = P.PRODUCT_ID
  8 GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9 HAVING AVG(R.RATING) >= 4
 10 ORDER BY AVERAGE_RATING DESC;
```

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
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202	Shirt	4.5
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201	Laptop	4.5
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PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
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